

Sina Sajadmanesh

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SUMMARY

Senior AI Engineer specializing in multimodal foundation models and video understanding, with a track record of leading end-to-end R&D and translating research into efficient, production-ready ML systems across heterogeneous hardware.

EDUCATION

Swiss Federal Institute of Technology (EPFL) , PhD in Electrical Engineering	Aug 2023
Sharif University of Technology , MSc in Information Technology Engineering	Sept 2016
University of Isfahan , BSc in Computer Software Engineering	Feb 2014

EXPERIENCE

Sony AI , Senior AI Engineer – Zurich, Switzerland	Jan 2026 – present
- Leading end-to-end R&D of a multimodal video understanding system using natural-language instructions to flexibly segment, retrieve, and annotate long-form video - Deployed video-language action recognition models on Apple Silicon for Sony's Remote Camera Tool, building an end-to-end Swift/Core ML pipeline with up to $53\times$ faster compute and $19\times$ lower peak memory - Developed compact vision-language models for image/video understanding and localization, achieving competitive accuracy at lower computational cost - Built agentic model-optimization workflows and a modular ML acceleration framework for deploying LLMs/VLMs and CV models across NVIDIA GPUs, Qualcomm NPUs, and Apple Silicon	
Sony AI , AI Engineer – Zurich, Switzerland	Oct 2023 – Dec 2025
- Developed a geometric optimization method for low-rank adaptation, resulting in a NeurIPS 2025 Spotlight paper - Led optimization of the Argus VFM inference stack across heterogeneous hardware, achieving up to $61\times$ speedup across 17 vision tasks - Led expansion of Argus into a unified 17-task vision foundation model, integrating 7 tasks including depth estimation, surface normals, super-resolution, and denoising - Architected a distributed multi-task learning platform unifying OpenMMLab libraries and powering training of Sony's Argus VFM	
Idiap Research Institute , Research Assistant – Martigny, Switzerland	May 2019 – Aug 2023
- Developed privacy-preserving machine learning methods for graph neural networks, with a focus on differential privacy - Published 3 papers at top-tier venues including CCS, USENIX Security, and WSDM, accumulating 340+ citations - Released 6 open-source research projects with 130+ GitHub stars and delivered 7 invited talks at leading universities and research institutions - Taught a short course on Trustworthy Machine Learning at the Artificial Intelligence Doctoral Academy	
Brave Software , Research Intern – Remote	Mar 2022 – May 2022
Developed a novel privacy-preserving federated learning framework for neural bandit models under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite	

SKILLS AND EXPERTISE

Foundation & Multimodal Models: Vision-Language Models · Video Understanding · ViT · CLIP · LLaVA · Large Language Models (LLaMA, GPT, Qwen) · PEFT (LoRA) · Self-Supervised Learning (MAE, DINO)

Model Optimization & Deployment: Core ML · MLX · TensorRT · ONNX · Qualcomm QNN / QAIIR · Quantization · Graph Optimization · Performance Profiling · Heterogeneous Hardware Deployment

Machine Learning Systems: PyTorch · Hugging Face (Transformers, PEFT, Accelerate, Datasets) · OpenMMLab (MMDetection, MMEval, MMDetection) · Distributed Training (DDP, FSDP)

Agentic AI & LLM Systems: LangChain · LangGraph · MCP · Tool Calling · RAG · Multi-Agent Systems · Workflow Orchestration

Engineering: Python · C++ · Bash · SQL · Docker · GitHub Actions · UV · Ruff

COMMUNITY AND PROFESSIONAL SERVICE

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), GNN User Group Meetup (2021), Twitter ML Seminar (2021)

Organizing / Program Committee: Privacy and Fairness in AI for Health (2023), PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2026, 2024), CVPR (2026), TIFS (2025), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

SELECTED PUBLICATIONS

StellA: Subspace Learning in Low-rank Adaptation using Stiefel Manifold Zhizhong Li, Sina Sajadmanesh, Jingtao Li, Lingjuan Lyu NeurIPS 2025 🌟 Spotlight (top 3%)	Dec 2025
Argus: A Compact and Versatile Foundation Model for Vision Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al. CVPR 2025	June 2025
ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees Sina Sajadmanesh, Daniel Gatica-Perez WSDM 2024	Mar 2024
GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, Daniel Gatica-Perez USENIX Security 2023	Aug 2023
Locally Private Graph Neural Networks Sina Sajadmanesh, Daniel Gatica-Perez CCS 2021	Nov 2021
A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics Seyed Ali Ossia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al. IEEE Internet of Things Journal	May 2020
Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al. WWW 2017	Apr 2017